

Pulkit Verma

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Research Interests

AI Safety, AI Planning, Action-Model Learning, Analysis of Abstractions, and Robotics.

Education

Ph.D. in Computer Science, Arizona State University

School of Computing and Augmented Intelligence

Advisor: [Siddharth Srivastava](#)

Thesis: Data Efficient Paradigms for Personalized Assessment of Taskable AI Systems

Tempe, USA

Exp. May 2024

Master of Technology, IIT Guwahati

Department of Computer Science & Engineering

Advisor: [Pradip K. Das](#)

Thesis: Resource Usage Analysis for Speech Recognition Techniques [↗](#)

Guwahati, India

May 2015

Bachelor of Engineering, SVITS, Indore

Information Technology

Advisor: [Anand Rajawat](#)

Capstone: Smart Analyzer - Predict and Cache Webpages based on Usage.

Indore, India

June 2011

Research and Professional Experience

Graduate Research Associate

AAIR Lab, Arizona State University

Advisor: [Siddharth Srivastava](#)

Research Areas: Agent Assessment, Model Learning, Automated Planning, Robotics

Tempe, USA

Aug. 2018 - (present)

Research Scientist Intern

Meta AI

Mentors: [Rohan Chitnis](#), [Alborz Geramifard](#), [Nitin Kamra](#)

Research Areas: Automated Planning, Curriculum Personalisation

Menlo Park, USA

May 2023 - Aug. 2023

Member of Technical Staff [Storage Efficiency Developer]

NetApp Inc.

Bengaluru, India

Jul. 2015 - Jul. 2018

Graduate Assistant

Robotics Lab, IIT Guwahati

Mentors: [Pradip K. Das](#), [Shivashankar B. Nair](#), [Shashi Shekhar Jha](#)

Research Areas: Speech Processing, Mobile Robotics

Guwahati, India

Dec. 2013 - May 2015

Assistant Systems Engineer [iOS Application Developer]

Tata Consultancy Services Ltd.

Pune, India

Mar. 2012 - Aug. 2013

Teaching Experience

Graduate-level Courses

Teaching Assistant

CSE 571 Artificial Intelligence, ASU

Spring'22, Fall'19

CS 566 Speech Processing, IIT Guwahati

Fall'14

Session Lead

CSE 571 Artificial Intelligence, ASU

Spring'23

CSE 574 Planning and Learning Methods in Artificial Intelligence, ASU

Spring'21

Undergraduate-level Courses

Teaching Assistant

CS 243 Software Engineering Lab, IIT Guwahati

Spring'15

CS 462 Computer Graphics Lab, IIT Guwahati

Spring'14

CS 513 Programming Lab, IIT Guwahati

Fall'13

Session Lead

CSE 471 Introduction to Artificial Intelligence, ASU

Fall'22, Fall'20

Honors & Awards

Graduate College Completion Fellowship, ASU Graduate College [↗](#)

June 2023

Highly competitive (1 per program per year) with tuition, stipend, and insurance for 1 year

Best Demo Award, AAMAS 2022 [↗](#)

May 2022

Awarded to one system demonstration at the AAMAS 2022 conference.

University (Engineering) Graduate Fellowship, ASU Engineering and Graduate College [↗](#)

March 2023

SCAI Doctoral Fellowship, School of Computing and Augmented Intelligence, ASU

March 2023

Spring 2022 GPSA Teaching Excellence Award, ASU GPSA [↗](#)

April 2022

SCAI Doctoral Fellowship, School of Computing and Augmented Intelligence, ASU

March 2022

Fulton Schools Experiential Learning Grant, ASU Engineering [↗](#)

Oct 2018

Post Graduation Fellowship, All India Council for Technical Education, Govt. of India [↗](#)

2013-2015

TCS ILP Kudos Award, Tata Consultancy Services Ltd. (Best Trainee)

May 2012

Conference Travel Awards

IJCAI-AIJ 2022 Travel and Accessibility Grant, IJCAI-ECAI 2022

July 2022

KR 2022 Student Grant, KR 2022

July 2022

ICAPS Student Scholarship, ICAPS 2022

May 2022

ASU SCAI Travel Award (for IJCAI-ECAI 2022, KR 2022, NeurIPS 2023)

2022-2023

ASU Graduate College Travel Award (for AAMAS 2021, ICLR 2021, ICML 2021,

2021-2023

IJCAI 2021, IJCAI-ECAI 2022, IJCAI 2023)

ASU GPSA Travel Grant (for AAAI 2021, AAAI 2022, IJCAI-ECAI 2022, KR 2022)

2021-2022

Publications

Conference Proceedings and Journals

[C1] **Pulkit Verma**, Rushang Karia, and Siddharth Srivastava. "Autonomous Capability Assessment of Black-Box Sequential Decision-Making Systems". In *37th Conference on Advances in Neural Information Processing Systems*, 2023. (To appear)

[C2] **Pulkit Verma**, Shashank Rao Marpally, and Siddharth Srivastava. "Discovering User-Interpretable Capabilities of Black-Box Planning Agents". In *19th International Conference on Principles of Knowledge Representation and Reasoning*, 2022. [↗](#)

- [C3] Naman Shah*, **Pulkit Verma***, Trevor Angle, and Siddharth Srivastava. “JEDAI: A System for Skill-Aligned Explainable Robot Planning”. In *21st International Conference on Autonomous Agents and MultiAgent Systems (Demonstration Track)*, 2022. **[Best Demo Award]**. [📄](#) [📺](#)
- [C4] Rashmeet Kaur Nayyar*, **Pulkit Verma***, and Siddharth Srivastava. “Differential Assessment of Black-Box AI Agents”. In *36th AAAI Conference on Artificial Intelligence*, 2022. [📄](#)
- [C5] Yizhong Wang et al. “Super-NaturalInstructions: Generalization via Declarative Instructions on 1600+ Tasks”. In *2022 Conference on Empirical Methods in Natural Language Processing*, 2022. [📄](#)
- [C6] **Pulkit Verma**, Shashank Rao Marpally, and Siddharth Srivastava. “Asking the Right Questions: Learning Interpretable Action Models Through Query Answering”. In *35th AAAI Conference on Artificial Intelligence*, 2021. [📄](#) [📺](#)
- [C7] **Pulkit Verma**, and Pradip K. Das. “A Comparative Study of Resource Usage for Speaker Recognition Techniques”. In *International Conference on Signal Processing and Communication*, 2016. [📄](#)
- [C8] **Pulkit Verma**, and Pradip K. Das. “i-Vectors in Speech Processing Applications: A Survey”. In *International Journal of Speech Technology*, 18(4), pp.529-546, 2015. [📄](#)
- [C9] Mayank Gupta, **Pulkit Verma**, Tuhin Bhattacharya, and Pradip K. Das. “A Mobile Agents based Distributed Speech Recognition Engine for Controlling Multiple Robots”. In *International Conference on Advances In Robotics*, 2015. [📄](#)
- [C10] **Pulkit Verma**, Mayank Gupta, Tuhin Bhattacharya, and Pradip K. Das. “Improving Services using Mobile Agents-based IoT in a Smart City”. In *International Conference on Contemporary Computing and Informatics*, 2014. [📄](#)

Workshop and Symposia Papers (Peer-Reviewed)

- [W1] **Pulkit Verma** and Siddharth Srivastava. “Learning Causal Models of Autonomous Agents using Interventions”. In *IJCAI 2021 Workshop on Generalization in Planning*, 2021. [📄](#)
- [W2] **Pulkit Verma**, Shashank Rao Marpally, and Siddharth Srivastava. “Learning User-Interpretable Descriptions of Black-Box AI System Capabilities”. In *ICAPS 2021 Workshop on Knowledge Engineering for Planning & Scheduling*, 2021. [📄](#)
- [W3] **Pulkit Verma** and Siddharth Srivastava. “Learning Interpretable Models for Black-Box Agents”. In *ICML 2020 Workshop on Human In the Loop Learning*, 2020. [📄](#)
- [W4] **Pulkit Verma**, and Siddharth Srivastava. “Learning Generalized Models by Interrogating Black-Box Autonomous Agents”. In *AAAI 2020 Workshop on Generalization in Planning*, 2020. [📄](#)
- [W5] Alok Shankar Mysore et al. “Investigating the ‘Wisdom of Crowds’ at Scale.”. In *Adjunct Proceedings of the 28th Annual ACM Symposium on User Interface Software & Technology (Poster)*, 2015. [📄](#)

Under Review

- [P1] **Pulkit Verma***, Rushang Karia*, Gaurav Vipat, Anmol Gupta, and Siddharth Srivastava. “Discovery and Assessment of Stochastic Capabilities for Sequential Decision-Making Systems”. 2023.
- [P2] Rushang Karia*, **Pulkit Verma***, Gaurav Vipat, and Siddharth Srivastava. “Epistemically Guided Exploration for Transferable Stochastic Planning in Non-stationary Relational Settings”. 2023.
- [P3] Daksh Dobhal*, Jayesh Nagpal*, **Pulkit Verma**, Rushang Karia, Rashmeet K. Nayyar, Naman Shah, and Siddharth Srivastava. “Using Explainable AI and Hierarchical Planning for Outreach with Robots”. 2023.

Doctoral Consortium

- [D1] **Pulkit Verma**. “Sample Efficient Paradigms for Personalized Assessment of Taskable AI Systems”. In 32nd International Joint Conference on Artificial Intelligence, 2023. [📄](#)
Mentor: [Peter Stone](#)
- [D2] **Pulkit Verma**. “Data Efficient Paradigms for Personalized Assessment of Taskable AI Systems – Dissertation Abstract”. In 32nd International Conference on Automated Planning and Scheduling, 2022. [📄](#)
Mentor: [Eva Onaindia](#)
- [D3] **Pulkit Verma**. “Data Efficient Paradigms for Personalized Assessment of Taskable AI Systems”. In 19th International Conference on Principles of Knowledge Representation and Reasoning, 2022.
Mentor: [Hector Geffner](#)
- [D4] **Pulkit Verma**. “Data Efficient Algorithms and Interpretability Requirements for Personalized Assessment of Taskable AI Systems”. In 30th International Joint Conference on Artificial Intelligence, 2021. [📄](#)
Mentor: [Leslie Pack Kaelbling](#)

Service

Leadership Roles

NeurIPS 2023 Workshop on Generalization in Planning (GenPlan'23) 🔗 [Lead Organizer]	2023
IJCAI 2022 Workshop on Generalization in Planning (GenPlan'22) 🔗 [Lead Organizer]	2022

Journal Reviewer

Journal of Artificial Intelligence Research (JAIR) 🔗	2023
IEEE Robotics and Automation Letters (RA-L) 🔗	2021

Conference PC Member

AAAI Conference on Artificial Intelligence (AAAI)	2023 🔗 , 2024 🔗
Intl. Conference on Autonomous Agents and Multi-Agent Systems (AAMAS)	2023 🔗 , 2024 🔗
International Conference on Learning Representations (ICLR)	2024 🔗
Advances in Neural Information Processing System (NeurIPS)	2023 🔗
International Joint Conference on Artificial Intelligence (IJCAI)	2023 🔗
International Conference on Automated Planning and Scheduling (ICAPS)	2022 🔗

Workshop PC Member

NeurIPS Workshop on Explainable AI in Action (XAIA)	2023 🔗
CoRL Workshop on Learning Effective Abstractions for Planning (LEAP)	2023 🔗
IJCAI Workshop on Bridging the Gap Between AI Planning and RL (PRL)	2023 🔗
ICAPS Workshop on Human-Aware and Explainable Planning (HAXP)	2023 🔗
CVPR W. on Open-Domain Reasoning Under Multi-Modal Settings (ODRUM)	2023 🔗
ICAPS Workshop on Explainable AI Planning (XAIP)	2021 🔗 , 2022 🔗
IJCAI Workshop on Generalization in Planning (GenPlan)	2021 🔗

Volunteer

International Joint Conference on Artificial Intelligence (IJCAI)	2021 🔗 , 2022 🔗
International Conference on Principles of Knowledge Representation & Reasoning (KR)	2018 🔗 , 2022 🔗
International Conference on Learning Representations (ICLR)	2021 🔗
International Conference on Automated Planning and Scheduling (ICAPS)	2021 🔗
Southwest Robotics Symposium	2019 🔗
Workshop on GPU Programming and Applications, IITG	2014 🔗

Other Academic Service

Member, (Inaugural) PhD Council, School of Computing and AI, ASU	2023-2024
Award and Grant Reviewer, Graduate and Professional Service Association (GPSA), ASU	2023

Mentorship

Masters Students

Gaurav Vipat , M.S. in Computer Science, ASU (co-authored [P1], [P2])	2023-(present)
Daksh Dobhal , M.S. in Computer Science, ASU (co-authored [P3])	2023-(present)
Jayesh Nagpal , M.S. in Robotics and Autonomous Systems, ASU (co-authored [P3])	2022-(present)
Trevor Angle , M.S. in Computer Science, ASU (co-authored [C3])	2021-2022
Shashank Rao Marpally , M.S. in Robotics and Autonomous Systems, ASU (co-authored [C2], [C6], [W2])	2020-2021

Undergraduate Students

Dylan Fulop , B.S. in Computer Science, ASU	2022-2023
Kyle J. Atkinson , B.S. in Computer Science, ASU	2021-2023
Namita Shah , B.S. in Computer Science, ASU	2022-2023
Khushi Patel , B.S. in Computer Science, ASU	2022-2023
Masih Tamsoy , B.Tech. in Computer Science and Engineering, IIT Guwahati	2014-2015

Other Affiliations

Graduate Student , Center for Human, AI, and Robot Teaming, ASU ↗	2021-(present)
Affiliated Graduate Student , Center for Human-Compatible AI, UCB ↗	2020-(present)
Member , Association for the Advancement of Artificial Intelligence (AAAI) ↗	2020-(present)
Graduate Student , Autonomous Agents and Intelligent Robots Lab, ASU ↗	2018-(present)
Participant , 3rd Summer School on Cognitive Robotics, USC ↗	July 2019
Online Contributor , Stanford Scholar Initiative ↗	2016
Graduate Student , Robotics Lab, IIT Guwahati ↗	2014-2015
